Puranjay Datta

Pre-Doctoral Researcher, Google DeepMind
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Education

Jul 2019 – Aug 2024 | **Indian Institute of Technology, Bombay** | **CPI: 9.41/10**
B.Tech. + M.Tech. in Electrical Engineering | *Dept. Rank 7/70*
Minor in Computer Science

Research Experience

Jul 2024 – Present | **Google DeepMind**
Pre-Doctoral Researcher | Advisors: Dr. Prateek Jain, Dr. Aditya Kusupati, Dr. Karthikeyan Shanmugam

Jul 2022 – Aug 2024 | **IIT Bombay**
Undergraduate Researcher | Advisors: Dr. Sharayu Moharir, Dr. Jaya Prakash Champati

May 2023 – July 2023 | **MITACS, University of Calgary**
Research Intern | Advisor: Dr. Hatem Abou-Zeid

May 2022 – July 2022 | **Texas Instruments**
Research Intern | Advisor: Jawaharlal Tangudu

Publications

S=In Submission, C=Conference, T=Technical Report, *=Equal Contribution

[C.3] Matryoshka Quantization
Puranjay Datta*, Pranav Nair*, Jeff Dean, Prateek Jain, Aditya Kusupati
International Conference on Machine Learning 2025
🏆 *Oral Sparsity in LLMs Workshop, International Conference on Learning Representations 2025* **[ICML'25]**

[C.2] Online Learning with Stochastically Partitioning Experts
Puranjay Datta, Sharayu Moharir, Jaya Prakash Champati
Uncertainty in Artificial Intelligence 2025 **[UAI'25]**

[C.1] Regret Bounds for Online Learning for Hierarchical Inference
Ghina Al-Atat, **Puranjay Datta**, Sharayu Moharir, Jaya Prakash Champati
Proceedings of the Twenty-fifth International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing **[MobiHoc'24]**

[T.2] Gemma 3n MatFormer Lab
Puranjay Datta, Aditya Kusupati, Ryan Mullins, Omar Sanseviero, Rakesh Shivanna, Gemma Team

[T.1] Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities
Gemini Team

[S.1] ROPES: Robotic Pose Estimation via Score-based Causal Representation Learning
Puranjay Datta*, Pranamya Kulkarni*, Emre Acartürk, Burak Varici, Karthikeyan Shanmugam, Ali Tajer
Embodied World Models for Decision Making Workshop, NeurIPS 2025
Under review at European Conference on Computer Vision 2026 **[In Submission]**

Selected Research Projects

Elastic Continued Pretraining
Advisors: Aditya Kusupati, Prateek Jain

→ Developed neuron reordering strategy for MatFormers, enabling efficient continual pretraining of smaller models.
→ Delivered models powering Gemini Embeddings & Google AI Overviews, with over 100,000 internal downloads.
→ Implemented Mix-n-Match for open-source Gemma 3n, allowing custom models via network parameter slicing.

Matryoshka Quantization
Advisors: Aditya Kusupati, Prateek Jain, Jeff Dean

→ Proposed a novel technique allowing a single trained model to be served at multiple precisions (e.g., int8, int4, int2).
→ MatQuant improved the int2 model quality by 4% on OmniQuant and 7% on QAT as base algorithms .
→ Enabled layer-wise Mix-n-Match, generating a combinatorial number of models at no additional training cost.

Anytime Thinking
Advisors: Aditya Kusupati, Prateek Jain

→ Developed a RL-training recipe to produce one single model for any thinking budget to control test-time compute.
→ Tested various natural language prompts and special token injection strategies to control thought to answer leakages.
→ Adaptive CoT outperforms fixed 8k budget baseline by 1.5% on Math and Code at 2k test-time thinking budget.

Value Function Training
Advisors: Arun Sai Suggala, Karthikeyan Shanmugam

→ Explored techniques to improve the reliability of value function predictions using long-range future awareness loss.
→ Implemented a Multi-Token Prediction (MTP) like configuration to mimic the TD(λ) learning for post-training.
→ Analyzed the effect of sequence length on value estimates and the role of high-entropy tokens in credit assignment.

Future Aware LLMs
Advisors: Sravanti Addepalli, Karthikeyan Shanmugam

→ Proposed distilling future tokens during training to improve at long reasoning tasks without extra inference compute.
→ Utilized the MatFormer architecture in a multi-task framework, where a harder future prediction objective on a larger model improves the performance of the core next-token prediction task of its sub-model.
→ Future token joint optimization improves upon isotoken MatFormer baseline with 0.01 gains in log perplexity.

Embodied World Model Understanding
Advisors: Karthikeyan Shanmugam, Ali Tajer

→ Developed an unsupervised Causal Representation Learning (CRL) framework to encode robot poses without labels.
→ Matched supervised baseline RoboPEPP's mse with much lower training compute in label-constrained settings.
→ Scaled theoretical CRL concepts using JAX by stabilizing score matching for high-dimensional robotics data.

Hierarchical Inference at Edge
Advisors: Sharayu Moharir, Jaya Prakash Champati

→ Explored optimal offloading strategy between small low-accuracy ML models on edge devices and large high-accuracy ML models on edge servers for inference tasks using EXP3 hedge online learning framework.
→ Designed Hedge-G & AdaHedge-G in the expert advice framework for dynamic IID experts under adversarial losses.
→ Proved Hedge-G achieves $\sqrt{T \log T}$ order-optimal regret, matching the theoretical lower bound for online experts.

Scholastic and Technical Accolades

→ Secured **All India Rank 132** in Kishor Vaigyanic Protsahan Yojana (KVPY) 2018
→ Secured **All India Rank 460** in **IIT JEE-Advanced** and **All India Rank 300** in **JEE Mains** 2019
→ Recipient of the **MITACS** Globalink Research Internship Fellowship 2023
→ **Received 2 Spot Bonuses** at Google DeepMind for delivering high impact models 2024
→ **One of 12, of 20k** applicants to be selected for the Google Pre-doctoral program 2024-26 2024
→ **Oral acceptance** at International Conference on Learning Representations Workshop on Sparsity in LLMs 2025

Teaching and Academic Service

→ TA for the Probability and Random Processes course at IIT Bombay.
→ Backend Lead for NPTEL Digital Signal Processing and its Applications : Addressed queries on an online forum and online YouTube doubt sessions, providing assistance to more than 5k enrolled students.
→ Volunteered for setting up inference servers for Gemini Models for pre-release product testing across diverse tasks.
→ Presented our work Matryoshka Quantization in the Google-wide Cool Papers Reading group.

Key Courses Undertaken

Machine Learning Probability and Random Processes,Stochastic Optimization, Convex Optimization, Markov Chains, Stochastic Control, Advanced Concentration Inequalities,Advanced ML, Intelligent & Learning Agents
Computer Science Data Structures and Algorithms, Logic for Computer Science, Design and Analysis of Algorithms, Game Theory and Algorithmic Mechanism Design, Computer Networks
Mathematics Calculus, Linear Algebra, Complex Analysis, Differential Equations, Number Theory and Cryptography